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19 September 1986
SUPERSEDING
MIL-H-81843B
22 March 1984

MILITARY SPECIFICATION
HOOD, FLYER'S, ANTI-EXPOSURE, INFLATABLE, HGU-32/P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type and size of an inflatable, anti-exposure, flyer's hood designated as HGU-32/P.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin, Gauge, Polyolefin
UU-P-268	-	Paper, Kraft, Untreated, Wrapping.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

MILITARY

MIL-C-5040	-	Cord, Nylon
MIL-I-6903	-	Ink, Marking (For Parachutes and Other Textile Items).
MIL-E-20652	-	Eyelets, Metallic, and Eyelet Washer, Metallic, General Specification for.
MIL-F-21840	-	Fastener Tape, Hook and Pile, Nylon.
MIL-C-83489	-	Cloth, Coated, Nylon, Polyurethane Coated.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center,Systems Engineering and Standardization Department,(SESD),(Code 93),Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC-N/A FSC 8475

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STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods.
FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking for U.S. Military Property.

2.1.2 Government Drawings,. The following Government Drawings form a part of this specifiction to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Air Systems Command

1639AS100	-	Hood Assembly, Flyer's Anti-Exposure, Inflatable, HGU-32/P
1639AS101	-	Hood Patterns, Flyer's Anti-Exposure, Inflatable, HGU-32/P

(Copies of specifications, standards,and government drawings, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein,(except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable law and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials and components. The materials and components shall conform to the applicable specifications and standards as listed or required herein (see 6.2.1h). Unless otherwise specified, the materials and components, used in the construction of the HGU-32/P, inflatable, anti-exposure, flyer's hood, shall have been manufactured not more than 12 months prior to the date of delivery of the HGU-32/P, inflatable, anti-exposure, flyer's hoods (see 6.2.1h).

3.2.1 Outer and innershells. The outer and innershells of the hood shall be fabricated from the polyurethane coated nylon cloth conforming to MIL-C-83489, type II. The color of the cloth shall match lusterless blue, number 35042, FED-STD-595.

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3.2.2 Oral valve. The heat sealable oral valve shall be as manufactured by the Halkey Roberts Corp., St. Petersburg, FL 33702-1098, Part Number 330-AC or equivalent. When the top portion is extended, the air shall be able to enter and escape from the air chamber of the hood through the valve into the atmosphere. The top portion, when depressed, shall seal off the flow of air through the valve and shall prevent entry of any kind of foreign matter.

3.2.3 Tape.

3.2.3.1 Seam. The seam tape shall be fabricated from the outer and innershell base cloth, 3.2.1, coated on both sides with 2.6 to 2.8 ounces per square yard of polyurethane. The coated cloth for the seam tape shall conform to the requirements of MIL-C-83489, except that the ounces per square yard shall be 6.9 to 7.5. The central portion of the hood shall be joined to each end portion by the seam tape, 3/4 ± 1/16 inch wide, as specified in Drawing 1639AS100. Each end of the seam tape shall be flush with the outer perimeter edge or shall not protrude beyond the edge of the hood by more than 1/8 inch.

3.2.3.2 Fastener. The hook and pile fastener tapes shall conform to MIL-F-21840, type I, class 1, 2 inches wide. The color shall approximately match the color of the outer and innershell cloth, 3.2.1, or may be black. Each hood shall contain one hook and one pile fastener tape. The length of the hook tape shall be 3 ± 1/8 inches and the length of the pile tape shall be 4 ± 1/8 inches.

3.2.4 Adhesive. The adhesive for cementing the fastener tapes to the cloth side of the outer and innershell cloth shall be Part Number X701, as manufactured by the Union Bay State Chemical Co., Cambridge, MA 02139, or equivalent. The adhesive shall be neutral in color or colorless.

3.2.5 Tie Cord. The tie cord shall be fabricated from the nylon cord conforming to MIL-C-5040, Type IA, color optional. The cut length of the cord shall be five feet plus or minus one inch. The cut ends of the cord shall be seared or treated to prevent fraying. No sharp edges shall be formed. The searing or treatment of the cut edges shall not damage the cord in any manner. One end shall be inserted through the eyelet and securely knotted to prevent loss. The remainder of the tie cord shall be faked into four inch hanks and held together at each end by the use of polychloroprene rubber bands.

3.3 Design. The hood shall consist of an outer and innershell heat sealed together to form an air chamber; an oral valve for orally inflating the air chamber with the capability to seal off the flow of air into the atmosphere and to prevent entry of any foreign matter into the air chamber; a fastener tape closure; A tie cord; and a pile tape for strobe light attachment. The hood shall conform in appearance to Drawing 1639AS100.

3.4 Construction. The construction of the hood shall be accomplished by the use of heat sealed seams and circular and/or bar bonded areas for flexibility and contour shaping of the hood when inflated. The heat sealed seam shall be straight, continuous, and parallel to the edge of the hood. The hood shall not contain any stitching. The coated side of the cloth shall be on the innerside of the hood air chamber as specified in Drawing 1639AS100. The hood shall be constructed in accordance with Drawing 1639AS100.

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3.4.1 Cutting. The inner and outershells shall be cut in strict accordance with Drawing 1639AS101 patterns furnished by the Government (see 6.4) which show location of the heat sealed area, fastener tapes, and oral valve, and markings for the proper assembly of the hood. The Government patterns shall not be altered in any manner and shall be used for making the working patterns. The working patterns shall be identical to the Government furnished patterns.

3.4.2 Shade marking. The outer and innershells shall be marked and ticketed, by any commercial method or bundled with the top ply of each individual component part marked with the bundle number, to assure the proper assembly of the hood. No metal fastening device or sewn-on ticket shall be used. The shade markings shall be accomplished by the use of an ink pad numbering machine, rubber stamp, or pencil, provided the markings do not show through to the outerside of the hood and are not deleterious to the cloth. The markings which are placed on the seam allowance shall not be visible on the outerside. The adhesive type shade marking tickets may be used for the shade markings (see 3.4.2.1). The assembled hoods shall not contain any shade marking tickets.

3.4.2.1 Adhesive shade marking tickets. The adhesive shade marking tickets shall be fabricated from paper with a thermo-activated adhesive applied on one side. The adhesive shall not discolor or damage the cloth and the adhesive mass shall not adhere to the cloth, upon removal of the ticket. The heat used to attach the thermo-activated adhesive tickets shall not stiffen, harden, scorch, or damage the cloth in any manner.

3.4.3 Attaching of seam tapes. The central vertical cut edges of the outer and innershell cloth, as applicable, shall be butted and sealed on the innerside with the seam tape (see 3.2.3.1). The seam tapes shall be heat sealed to the seams without tension so that an equal amount is on each side of the seam edge which it covers as shown in Drawing 1639AS100. Each end of the seam tape shall be further bonded to each individual shell with an approximately 3/4 inch diameter bond.

3.4.4 Use of adhesive. In the cementing of the fastener tapes, the surface, to which the adhesive is to be applied, shall be thoroughly cleaned with a suitable solvent so that the dusting materials (zinc stearate or talc) and any other surface contaminant shall be removed. The surface shall be clean, prior to the cementing of the fastener tapes. Care shall be exercised to insure that the coating and the base undercloth shall not be damaged and that the adhesion between both shall not be impaired in any manner by prolonged exposure to the solvent. The solvent used shall evaporate completely, prior to the application of the adhesive, and shall leave no residue. The cemented areas shall not contain any trapped air, channel or wrinkle. The adhesive shall be controlled to insure that old adhesive or adhesive which has partly or completely polymerized shall not be used. The containers for the adhesive shall be free from congealed adhesive before being refilled.

3.4.5 Cementing of the fastener tapes. The hook and pile fastener tapes shall be located on the hood as specified in the patterns (see 3.2.3.2). The back side of each fastener tape shall be cemented to the appropriate outer or innershell hood cloth in the manner shown in Drawing 1639AS100. Use of the

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adhesive shall be as outlined in 3.4.4. The adhesive shall be applied in a straight line, parallel to the edges of the tapes. Each tape shall measure 2 by 3 inches. The cured adhesive shall present a neat and uniform appearance. In addition, the cured adhesive shall not cause the hood to be uncomfortable to the user during wearing, donning or doffing. The adhesive, upon curing, shall not cause the cloth or fastener tapes to shrink or pucker at any point on the hood. The hood shall be free from any congealed mass of adhesive and any spot or stain resulting from excess adhesive. A minimum amount of zinc stearate or talc shall be applied to the cloth side of the hood's outer and innershell cloth to prevent unwanted adherence of surfaces. The area (outer and innershell cloth) surrounding the hook fastener tape shall be heat sealed together as specified in Drawing 1639AS100.

3.4.6 Metal Eyelet. One metal eyelet, in accordance with MIL-E-20652, Part Number M20652/1-BNE109, shall be installed in the center of a circular heat sealed bond located in the center of the outershell, Part Number 1639AS101-5; and the innershell, Part Number 1639AS101-6, respectively. The metal eyelet shall accommodate the tie cord specified in 3.2.5.

3.5 Performance requirements

3.5.1 Seam adhesion. The seam adhesion of all heat sealed areas, regardless of their configurations (circular and/or straight), shall be 12 pounds or more.

3.5.1.1 Outer perimeter. The seam adhesion of the outer perimeter which joins the outershell to the innershell of the hood, when inspected as specified in 4.6.1.1, shall be not less than 12 pounds.

3.5.1.2 Taped seams. The seam adhesion of each taped seam of the outer and innershells, which joins the central portion of the hood to each side portion, when inspected as specified in 4.6.1.2, shall be not less than 12 pounds.

3.5.2 Leakage. When inspected as specified in 4.6.3, the pressure in the hood shall not drop below 1.5 inches (water column). The seams, bar bonded areas and the base of the oral valve shall remain perfectly intact and shall show no indication of separation. There shall be no evidence of constructional or material failure in any respect. There shall be no evidence that the air entering the hood air chamber is hindered or that the front or back of the oral valve is blocked by talc, zinc stearate, or other foreign matter. There shall be no evidence of any loss of air from the hood air chamber, when the top portion is depressed fully in the oral valve. When the top portion of the oral valve is fully extended, it shall permit the air to flow through the valve into the atmosphere. The flow of the air shall not be blocked or restricted by any excess adhesive, talc, zinc stearate or other foreign matter. The attachment of the base of the oral valve to the hood shall remain perfectly intact.

3.6 Markings. The markings on the hood shall be in accordance with MIL-STD-130 and Drawing 1639AS100. Ink, color, size and location of markings shall be as specified in Drawing 1639AS100. All markings shall be thoroughly dry prior to packaging and packing.

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3.6.1 Serial Numbers. Each hood will be identified by a serial number located as specified in Dwg 1639AS100. Markings shall be in accordance with 3.6. Serial numbers shall be assigned by the manufacturer and shall consist of a block of consecutive numbers to cover the entire acquisition quantity.

3.7 Workmanship. The hoods shall be thoroughly cleaned and all loose fiber, lint, and foreign matter shall be removed. The hoods shall not contain any spot, stain, nonspecified hole, abraded area, tear, cut, mend or patch. Because of the emergency and life support use of these hoods, the importance of providing a product of uniform excellent quality cannot be overemphasized. The hoods shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability or durability. The hoods shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection - First article inspection shall consist of examinations and tests performed on samples which are representative of the production item after the award of a contract to determine that the production item meets the requirements of this specification (see 4.3).

b. Quality conformance inspection - Quality conformance inspection shall consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.4).

4.3 First article inspection. The first article inspection of the HGU-32/P, inflatable, anti-exposure, flyer's hoods shall consist of examinations and tests for all of the requirements of this specification.

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4.3.1 First article samples. Unless otherwise specified in the contract, as soon as practicable after the award of the contract or purchase order, the manufacturer shall submit three hoods. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these hoods, from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2.1f). Approval of the first article inspection samples or the waiving of the first article inspection does not waive the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2.1g).

4.3.2 First article sample approval. Upon completion of the first article inspection program, the Government activity responsible for coordinating the inspection program (see 6.2.1g) shall report to the Contracting Officer the results of the inspection program with appropriate recommendations. One of the approved first article inspection samples of the hoods will be returned to the manufacturer for use in monitoring production. The other two hoods will be destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under the contract or purchase order.

4.4 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

a. Visual examinations of the assembled hoods
b. Leakage
c. Packaging

4.4.1 Sampling.

4.4.1.1 Inspection lot.

4.4.1.1.1 Hoods. An inspection lot size shall be expressed in units of one hood made essentially under the same conditions and from the same materials and components. The sample unit shall be one hood.

4.4.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing hoods, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing hoods, fully prepared for delivery with the exception that it need not be sealed.

4.4.1.2 Sampling for tests and examinations of the hoods and packaging. The sample size, acceptance criteria, tests and examinations required for the hoods or the packaging, as applicable, shall be as specified in table I.

4.5 Inspection conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified in the contract, all the inspections required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77° ± 18°F (25° ± 10°C). When the inspections are conducted at an atmospheric pressure or temperature different from the above values, proper correction shall be made for the change in instrument reading.

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4.5.2 Inspection area and equipment. The area in which the hoods are inspected shall be adequately protected to preclude damage to the units. The area and inspection equipment shall be free of sharp or rough edges, burrs, protrusions, etc., which will cut, tear or damage the hoods or the components.

4.5.3 Air. When use of air is specified in an inspection, the air used shall not contain any oil or liquid water.

4.6 Inspection methods.

4.6.1 Seam adhesion.

4.6.1.1 Outer perimeter. Five specimens, 1 ± 1/16 inch wide and 6 ± 1/16 inches long, shall be cut across and perpendicular to the seam of the outer perimeter (outershell bonded to the innershell) of the hood. The specimens shall be cut from different locations so as to be representative of the sealing of the outer perimeter seam. The specimens shall be taken from the straight portion of the outer perimeter of the hood. One free end of one specimen (end not bonded) shall be placed in the upper jaw and the other free end in the lower jaw of a suitable inspection apparatus equipped with an autographic recording device (see FED-STD-191, Method 5100). The seam portion of the specimen shall be in the center between the two jaws. The jaws shall be separated until failure occurs, either breaking of the cloth or separation of the bond. In either case, the result obtained shall be not less than that specified in 3.5.1.1. The highest value obtained shall be taken as the result for the outer perimeter seam adhesion for the individual inspected specimen. The rate of jaw separation under no load shall be 12 ± 1/2 inches per minute. The front clamp of each jaw shall be 1 by 3 inches. The remaining four specimens shall be similarly treated and inspected. The results of the five inspected specimens shall not be averaged. Each individual specimen shall conform to the outer perimeter seam adhesion requirement specified in 3.5.1.1.

4.6.1.2 Taped seams. Three specimens, 1 ± 1/16 inch wide and 3 ± 1/16 inches long, shall be cut across and perpendicular to one of the taped seams of the outershell of the hood. Three additional specimens shall be similarly cut to determine the seam adhesion of the other taped seam of the outershell. Specimens shall also be similarly cut to determine the seam adhesion of the taped seams of the innershell. Each set of the three specimens shall be marked for identification as to location. The specimens shall be cut from different locations so as to be representative of the sealing of the outer and innershell taped seams. The specimens shall be taken from the straight portion of the taped seams of the outer and innershell of the hood, as applicable. One free end of one specimen (end not bonded) shall be placed in the upper jaw and the other free end in the lower jaw of a suitable inspection apparatus equipped with an autographic recording device (see FED-STD-191, Method 5100). The seam portion of the specimen shall be in the center between the two jaws. The jaws shall be separated until failure occurs, either breaking of the cloth or separation of the bond. In either case, the result obtained shall be not less than that specified in 3.5.1.2. The highest value obtained shall be taken as the result for the taped seam adhesion for the individual inspected specimen. The rate of jaw separation under no load shall be 12 ± 1/2 inches per minute. The front clamp of each jaw shall be 1 by 3 inches. The remaining two specimens shall be similarly treated and inspected. The inspection results of the three specimens of each set shall not be averaged. Each individual specimen shall conform to the taped seam adhesion requirement specified in 3.5.1.2.

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4.6.2 Visual examination.

4.6.2.1 Hoods. Every hood shall be examined visually for major defects to determine conformance to this specification. Each hood, selected as a sample unit from the lot, shall be examined visually for minor defects to determine conformance to this specification. The classification of defects, table II, shall be used to classify the defects found.

4.6.2.2 Packaging. Each of the fully prepared shipping containers, containing hoods, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing and marking conform to this specification. The list of defects, table III, shall be used to enumerate the defects found.

4.6.3 Leakage. After not less than 24 hours has elapsed from the thermal bonding of the seams, bar areas and oral valve, the hood shall be inflated with air (see 4.5.3) to a pressure of 2 inches (water column). The air supply shall be completely shut off and after not less than 5 minutes, the pressure in the hood shall be checked and readjusted, if necessary, to the original pressure of 2 inches (water column). At the end of not less than 5 minutes after the readjustment period, the pressure in the hood air chamber shall be measured (see 3.5.2). While inflated, the hood shall be examined for material and construction failure and separation of the seams, bar bonded areas and the attachment of the base of the oral valve to the hood air chamber (see 3.5.2). The air introduced into the hood through the oral valve shall be observed to determine whether the flow of the air through the oral valve is restricted or blocked by talc, zinc stearate or other foreign matter (see 3.5.2). Upon completion of the testing and examination, the method of introducing the air through the oral valve shall be removed from the oral valve and the top portion fully depressed. The valve shall be observed for leakage (see 3.5.2). Upon completion of the examination for leakage, the top portion shall be fully extended and the flow of the air through the oral valve into the atmosphere shall be observed (see 3.5.2).Upon completion of the examination for the release of the air from the hood air chamber, the attachment of the base of the valve to the hood shall be examined (see 3.5.2). Upon completion of the valve attachment examination, the hood air chamber shall be completely deflated.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A.

5.1.1 Level A. The air chamber of each hood shall be completely evacuated of air and the top portion of the oral valve shall be fully depressed. Each hood shall be folded as follows:

a. Lay the hood flat so that the outer perimeter edges coincide with each other.

b. Fold the hood in half, longitudinally.

c. Fold the hood in half, horizontally.

d. Care shall be exercised to assure that the oral valve on the innerside of the hood is not bent, distorted or deformed.

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Each hood shall then be placed into a snug fitting, clear, bag constructed of polyolefin material conforming to Federal Specification L-P-378, Type I, Class 4, Grade B, Finish 2, Thickness 0.00125 plus or minus 25 percent inch gage film. A label conforming to 5.3.1 shall be inserted into each bag, or the marking specified in 5.3.1 shall be printed directly on the bag. The heat sealed seams shall be straight, continuous and parallel to each other and the edges of the polyolefin bag. The closing heat sealed seam of the bag shall be as close as possible to the edge of the bag. No hood shall be deformed or distorted in the bagged condition nor shall the polyolefin bag be damaged. No part of any hood shall be caught in any heat sealed seam. One corner of each bag shall contain a hole, 1/4 inch in diameter, to allow the excess air to escape. As an alternate, the final closure of the bag may be accomplished by use of a tuck or reverse flap. Twelve bagged hoods shall be placed within a snug fitting fiberboard box conforming to PPP-B-636, Type CF or SF, domestic class, variety SW, grade 275. The top and bottom of the contents shall be completely covered with a sheet of 30 pound minimum basis weight kraft paper conforming to UU-P-268, Grade B. All the seams, corners and manufacturer's joint shall be sealed with tape not less than 3 inches wide conforming to PPP-T-45, Type III, Class 2, Grade A, B or C. The container shall be constructed and closed in accordance with the appendix to PPP-B-636. The body joint and the top and bottom flaps shall be firmly glued together as specified in PPP-B-636. The fiberboard container shall not contain any metal fastenings or stitches.

5.2 Packing. Packing shall be level A, B or industrial, as specified (see 6.2.1i).

5.2.1 Level A. Eight boxes, 96 hoods, packaged as specified in 5.1, shall be packed as specified in 5.2.2, except that the fiberboard container shall be weather-resistant Class, Grade V2s. The container shall be constructed, closed, waterproofed and reinforced in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Eight boxes, 96 hoods, packaged as specified in 5.1, shall be packed within a snug fitting fiberboard container conforming to PPP-B-636, Type CF or SF, Domestic class, Variety SW, Grade 275. Each shipping container shall be constructed and closed in accordance with method II as specified in the appendix to PPP-B-636.

5.2.3 Industrial. The packaged hoods, which require packing for acceptance by the carrier, shall be packed within exterior type shipping containers in a manner that shall insure safe transportation, at the lowest rate, to the point of delivery. The containers shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2.1j), the interior and exterior fiberboard containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year).

5.3.1 Label. A white paper label, with legible and durable black letters and numerals, shall be placed inside each polyolefin bag. The label shall be placed so that the information on the label shall be capable of being read through the bag. If desired, instead of a white paper label, the markings may be imprinted directly onto the outerside of the bag. The paper label or polyolefin bag, as applicable, shall contain the following information:

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a. Government part number
b. National Stock Number
c. Nomenclature
d. Quantity
e. Date of Manufacture (month and year)

6. NOTES

6.1 Intended use. The HGU-32/P, inflatable, anti-exposure, flyer's hood covered by this specification is intended to be used to protect the head of the wearer from exposure to cold water, wind and spray as a result of emergency ditching of an aircraft at sea. The hood is intended to be used with anti-exposure coveralls.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification including any amendments.

b. Applicable Government drawings, including revisions.

c. Government part number.

d. National stock number.

e. Quantity desired.

f. Whether first article inspection is waived (see 4.3.1).

g. Name and address of the first article inspection laboratory (see 4.3.1) and the name and address of the Government activity responsible for coordinating the first article inspection program (see 4.3.2)

h. Certificate of compliance to be provided by the contractor for:

(1) Materials and components conforming to applicable specifications and standards (see 3.2)

(2) The age of materials (see 3.2 and 6.2.2)

i. Preservation requirements (see 5.1) and applicable level of packing (see 5.2).

j. Whether any special markings are required (see 5.3)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the

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provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.	Options
(a) 3.3	Certificate of compliance for the age of materials and components.	DI-E-2121	------
(b) 4.3.2	First article inspection reports.	DI-T-5329	------

(Copies of data item descriptions required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the items should be a first production item. The first article should consist of three units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Patterns. The patterns for the hood will be furnished by the contracting officer to the contractor for use in cutting working patterns (see 3.4.1).

6.5 Subject term (keyword) listing.

Oral
Tie cord
Anti-exposure

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - AS

Review activities:
Army - AV

Preparing activity:
Navy - AS

(Project No. 8475-0202)

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TABLE I. Sample size, acceptance criteria, tests and examinations of the hoods and packaging.

Inspection	Paragraph		Sample Size	Acceptance Criteria
	Requirment	Method		
Visual Examination	3.3 through 3.4.6, 3.6, 3.7 and Drawing 1639AS100	4.6.2.1 and Table II	a). Every hood for major defects b). Inspection Level II 2/ for minor defects	a). Reject all units with any major defects. b). An Acceptable Quality Level of 6.5 defects per 100 units for minor defects.
Leakage 1/	3.5.2	4.6.3	Every hood	Reject all defective units
Packaging	Section 5	4.6.2.2 and Table III	Inspection Level S-2,2/	An Acceptable Quality Level of 2.5 defects per 100 units

1/ The results of the leakage inspection shall be identifiable by the assigned serial numbers which shall be marked on the hoods as specified in 3.6.1.

2/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

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TABLE II. Classification of defects for the visual examination of the hoods.

Defect	Major	Minor
GENERAL		
a. Any non-specified hole (including pin hole), scissors or knife cut, tear, burn, mend, patch, or weakening defect such as smash, multiple float, loose slub or abraded area	X	
b. Any portion of the cloth stiffened or hardened by any process of manufacturing	X	
c. Color of any component not specified, not uniform, any part shaded or any shade bar		X
d. Coated side of the cloth on the outerside of the hood	X	
e. Any spot or stain, any shade ticket not removed, excluding the metal fastening device or sewn-on kind, or any shade identification stamping visible on the outerside of the hood		X
f. Any metal fastening device or sewn-on shade marking ticket attached to the hood	X	
g. Any portion of any heat sealed seam or bonded area open	X	
h. Any heat sealed seam twisted, puckered, or not straight, continuous or parallel to the edge of the hood or the seam or covering tape edge, as applicable	1/	
i. Any trapped air, wrinkle or crease in the area of the oral valve		X
j. Base of any oral valve heat sealed to the cloth side of the innershell cloth	X	
k. Any oral valve not located on the left innerside of the hood	X	
l. Any taped seam edge of the central portion or of any side portion overlapped more than 1/16 inch or open more than 1/16 inch	1/	
m. Any 3/4 inch diameter bond missing at end of any seam tape.	X	

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TABLE II. Classification of defects for the visual examination of the hoods. - Continued

Defect	Major	Minor
FASTENER TAPE		
a. 10 to 25 percent of the hooks or pile flattened or missing from any tape		X
b. 25 percent or more of the hooks or pile flattened or missing from any tape	X	
c. Any not specified size		X
d. Any misplaced not serving the intended purpose	X	
e. Any area of the hook or pile tape, one inch or more in length, that contains fiber scraps, lint, cement or other impurities		X
f. The attachment of any fastener tape twisted, puckered, pleated or any part of the hood or any fastener tape caught in an unrelated operation		X
COMPONENTS AND ASSEMBLY		
a. Any component not as specified or any defect of a component or defect of assembly not herein classified	1/	
b. Any component, component part, or required operation omitted, or any operation improperly performed not herein classified	1/	
c. Any component part not cut in accordance with the patterns	1/	
MARKINGS		
a. Missing, incomplete, incorrect, illegible or improperly located		X

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TABLE II. Classification of defects for the visual examination of the hoods. - Continued

Defect	Major	Minor
METAL EYELET		
a. Missing or improperly located	X	
b. Not as specified	X	
c. Not securely clinched in material	X	
d. Contains sharp edges, cracks or protrusions	X	
e. Inner or outer diameter out of round by more than 1/16 inch		X

1/ The defect shall be classified as major, when it seriously affects the appearance or serviceability, otherwise it shall be classified as a minor defect.

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TABLE III. List of defects for packaging.

Item	Defect
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence or method of application; markings not the same on the interior and exterior containers; information on the label on the polyolefin bag not readable through the film, no label in the polyolefin bag, or no markings on the polyolefin bag, as applicable.
Materials	Any non-conforming component; any component missing, damaged or otherwise defective.
Workmanship	Inadequate application of the components, such as incomplete closure of the container flap or loose strapping; bulging or distortion of any container; any interior fiberboard container containing any metal fastener or stitches; any portion of any hood distorted or caught in any heat sealed seam; any pololefin bag damaged, any vent hole missing, or any open or non-continuous heat sealed seam.
Content	Number per container is more or less than required; gross or net weight exceeds the requirements; any hood not folded as specified; any hood air chamber not completely evacuated of air; any top portion not fully inserted into the oral valve.

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